

Grade 7 Accelerated  
Unit 3  
Lesson 4 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Write an equivalent exponential expression by applying the laws of exponents.

1.  $(2^6)^2 = \underline{\hspace{2cm}}$

2.  $11^1 \cdot 11^7 = \underline{\hspace{2cm}}$

3.  $\left(\frac{-3}{10}\right)^4 = \underline{\hspace{2cm}}$

4.  $(6 \cdot -8)^3 = \underline{\hspace{2cm}}$

Rewrite each expression as an equivalent exponential expression using positive exponents.

5.  $\frac{1}{-5^{-5}} = \underline{\hspace{2cm}}$

6.  $(-3)^{-9} = \underline{\hspace{2cm}}$

7.  $3^{-4} = \underline{\hspace{2cm}}$

8.  $\frac{3^{-7}}{12^{-1}} = \underline{\hspace{2cm}}$



9. For each expression, rewrite the exponential expression applying the negative exponent law, writing in expanded form, applying the product of powers exponent law and evaluate.

| Exponential Expression   | Apply Negative Exponent Law | Expanded Form | Apply Product of Powers Exponent Law | Value |
|--------------------------|-----------------------------|---------------|--------------------------------------|-------|
| $(-3)^2 \cdot (-3)^{-5}$ |                             |               |                                      |       |
| $2^8 \cdot 2^{-4}$       |                             |               |                                      |       |
| $9^9 \cdot 9^{-9}$       |                             |               |                                      |       |

10. For each expression, rewrite the exponential expression applying the negative exponent law, writing in expanded form, applying the quotient of powers exponent law and evaluate.

| Exponential Expression     | Apply Negative Exponent Law | Expanded Form | Apply Quotient of Powers Exponent Law | Value |
|----------------------------|-----------------------------|---------------|---------------------------------------|-------|
| $(-5)^{-6} \div (-5)^{-4}$ |                             |               |                                       |       |
| $2^0 \div 2^{-3}$          |                             |               |                                       |       |
| $3^{-7} \div 3^5$          |                             |               |                                       |       |